15 STRINGS IN PYTHON

Learning Outcomes

The students will be able to:

- create strings.
- learn to access the characters of strings.
- know various operators of strings.
- understand the concept of escape sequences.
- learn various string functions.

A string consists of a sequence of characters. String characters include letters, digits, or symbols. Strings are used to represent textual. The purpose of string is to store human understandable sentences. A string may have a single word or collections of words or symbols.

Some examples of strings are:

- Hello! How are you?
- In which class are you studying?
- India

In Python, a string can be enclosed in single (') or double ("") quotes. Strings can be manipulated using operators and string functions.

Creating Strings

Creating strings in Python is similar to assigning a value to a variable. For example, S="HELLO WORLD". A string can have spaces in between the words. A string may consist of different types of data values. Each data value in the string is called an element of the string. It is identified with a unique index value. In Python, these index numbers are used for accessing elements of the string. In a string, all the elements can be different from one another or there can be identical elements.

Index Value

0	1	2	3	4	5	6	7	8	9	10
H	E	L	L	O		W	O	R	L	D

In the given description, there are similar elements in the string. For example, the letter 'L' is repeated thrice and each 'L' has a unique index value.

Accessing String elements using index value:

Every string is stored in the computer memory as a sequence of characters. Each character in the string has an index value, which denotes its position within the string. The first character has an index 0, the second character has an index 1, the third character has an index 2, and so on. This type of indexing is called zero-based indexing. Index can be used for accessing individual elements

of the string. The index value can be negative also. The negative index starts from -1 and is assigned from right to left.

a = "PYTHON WORLD"

0	1	2	3	4	5	6	7	8	9	10	11
P	Y	T	H	O	N		W	O	R	L	D
-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1

Activity 15.1

Write the output of the print command in the given table.

a = "Python World"

Print a[0]	
Print a[-2]	
Print a[5]	
Print a[7]	
Print a[-9]	
Print a[10]	

Traversing a String

Traversing a string means to access its each character. A 'for' or 'while' loop can be used for this purpose.

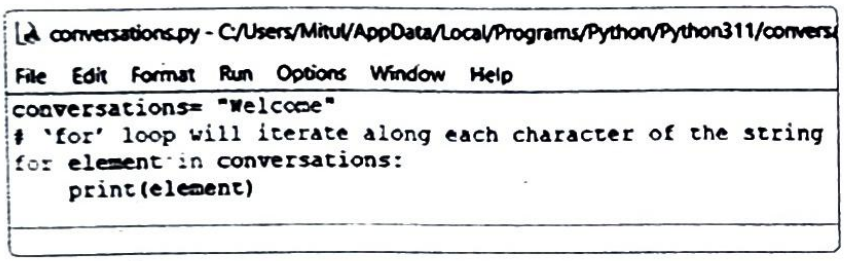
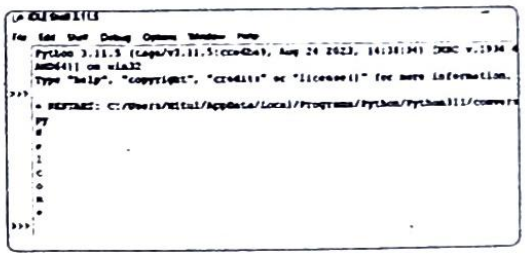


Figure 15.1: Traversing the String using the 'for' Loop



Output

Multiline Strings

In python triple quotes are used to declare a multiline string.

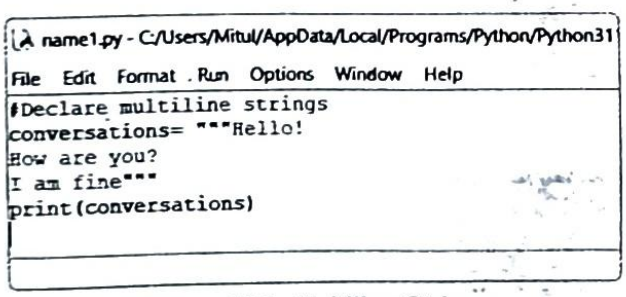
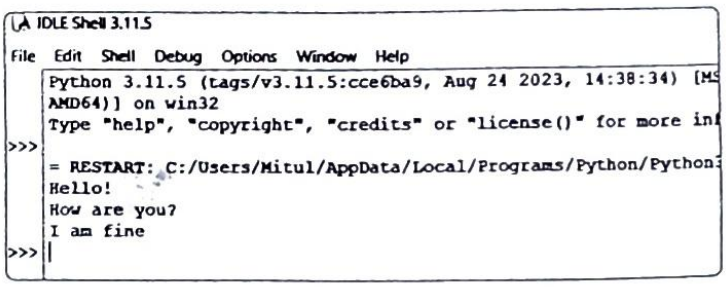


Figure 15.2: Multiline Strings



Output

Concatenating String

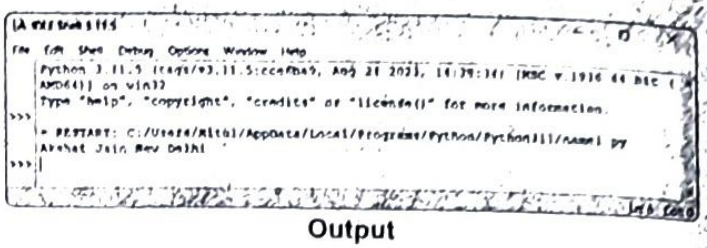
Concatenating means to combine two or more strings. In python, the '+' operator is used to concatenate strings.

name1.py - C:/Users/Mitul/AppData/Local/Programs/Python

File Edit Format Run Options Window Help

```
Name = "Akshat "
Surname = "Jain "
Address = "New Delhi "
concat_string = Name+Surname+Address
print (concat_string)
```

Figure 15.3: Concatenating Three Strings



Output

Escape Sequences

To insert characters that are not allowed in a string, an escape character is used. Backslash (\) is considered as escape character. The character that you want to insert should be written after backslash. The combination of escape character (\) and the character that you want to written is called as escape sequence. The following are the examples of escape sequences:

Character	Purpose	Example

\'	To insert single quotes in a string	`str="it\'s my pen"`
		`print(str)`
		Output
		`it's my pen`

\n	To insert new line in a string	`print("It\'s my pen \n I like it very much")`
		`print(str)`
		Output
		`It's my pen`
		`I like it very much`

\\	To insert backslash in a string	`inform="The symbol of backslash is \\"`
		`print(inform)`
		Output
		`The symbol of backslash is \`

\t	To provide space for one tab	`str="This pen is of which brand?\t Pentonic"`
		`print(str)`
		Output
		`This pen is of which brand?  Pentonic`

Activity 15.2

Create your portfolio, which should include your name, parent's name, class and address using strings in Python.

String Operators

In Python, many operators exist that have some specific meaning for strings. The following table shows the Python operators that can be used with strings:

Operator	Purpose	Example

Replication operator: *	The replication operator in Python is used to repeat a string for a specified number of times.	`s1="Hello"`
		`print(s1*4)`
		Output: `HelloHelloHelloHello`

Concatenation operator: +	The concatenation operator(+) is used for joining multiple strings.	`s1="Hello! "`
		`s2="Hi "`
		`s3="How are you?"`
		`print(s1+s2+s3)`
		Output: `Hello! Hi How are you?`

Slicing operator: []	Provides the character from the given index	`a="India"`
		`prit(a[2])`
		Output: `d`

Range slice operator: [:]	Provides characters from the given range	`a="Python"`
		`print(a[2:4])`
		Output: `th`

Membership operator: in	Returns true if a character exists in the given string	`str="Mountain"`
		`if 't' in str:`
		`  print("t is present in mountain")`
		`else:`
		`  print("t is not present in mountain")`
		Output: `t is present in mountain`

Membership operator: not in	Returns true if a character does not exist in the given string	`str="Mountain"`
		`if 'k' not in str:`
		`  print("k is not present in mountain")`
		`else:`
		`  print("k is present in mountain")`
		Output: `k is not present in mountain`

Activity 15.3

What would be the output of following program?

```
print("Hello\nWorld!")
print("Python\tProgramming")
```

String In-built Functions

The following are some examples of the in-built functions of String:

len(): This function is used to find the length of a string. It calculates the number of elements in the string and returns an integer value.

Syntax:

```
int len(string)
```

```
name1.py - C:/Users/Mitul/AppData/Local/Programs/Python/Python311/name1.py (3.11
Edit Format Run Options Window Help
a="Hello India"
lenght = len(a)
print("Lenght of string is: ", lenght)
```

Figure 15.4: **Using the len() Function**

```
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24
AMD64)] on win32
Type "help", "copyright", "credits" or "lic
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Pro
Lenght of string is:  11
>>>
```

Output

upper(): This function converts all the characters of the input string into uppercase.

Syntax:

```
string string.upper()
```

```
name1.py - C:/Users/Mitul/AppData/Local/Programs/Python/Python311/name1.p
Edit Format Run Options Window Help
a="Hello India ! "
a_upper=a.upper()
print("String in capital letters : ",a_upper)
```

Figure 15.5: **Using the upper() Function**

```
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 202
AMD64)] on win32
Type "help", "copyright", "credits" or "license
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Program
String in capital letters :  HELLO INDIA !
>>>
```

Output

lower(): This function converts all the characters of the input string from uppercase to lowercase.

Syntax:

```
string string.lower()
```

```
name1.py - C:/Users/Mitul/AppData/Local/Programs/Python/Python311/na
File Edit Format Run Options Window Help
a="Hello India ! "
a_lower_case=a.lower()
print("String in small letters : ",a_lower_case)
```

Figure 15.6: **Using the lower() Function**

```
File Edit Shell Debug Options Window Help
Type help, copyright, credits or license()
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Programs
String in capital letters :  HELLO INDIA !
>>>
=== RESTART: C:/Users/Mitul/AppData/Local/Progra
String in small letters :  hello india !
>>>
```

Output

endswith(): This function returns True if the string ends with a particular suffix.

Syntax:

```
boolean string.endswith(value)
```

```
File Edit Format Run Options Window Help
data = "Hari is a tall boy"
val = data.endswith("boy")
print("Boy is the last word in string: ",val)
```

Figure 15.7: **Using endswith() Function**

```
File Edit Shell Debug Options Window Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 1
AMD64)] on win32
Type "help", "copyright", "credits" or "license()"
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Programs/Py
Boy is the last word in string:  True
>>>
```

Output

Activity 15.4

What would be the output of following program?

```
str2="My classmate is Aayan"
if str2.endswith("Aayan"):
    print("Aayan is studying in class X")
else:
    print("Aayan is not the student of class X")
```

_____

_____

find(): It finds the first occurrence of the specified value. It returns -1 if the value is not found.

Syntax:

```
int string.find(value)
```

```
File  Edit  Format  Run  Options  Window  Help
information="Jerry is studying in class X !"
val=information.find("i")
print("The index of i is: ", val)
```

Figure 15.8: Using the find() Function

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2
AMD64)] on win32
Type "help", "copyright", "credits" or "licen
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Progr
The index of i is:  6
>>>
```

Output

islower(): This function returns True if the string is in lower case.

Syntax:

```
boolean string.islower()
```

```
File  Edit  Format  Run  Options  Window  Help
information="hello india"
val=information.islower()
print("The given string is in lower case: ", val)
```

Figure 15.9: Using the islower() Function

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" fo
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Programs/Pyth
The given string is in lower case:  True
>>>
```

Output

replace(): This function is used to replace a character or word with a new character or word in string

Syntax:

```
string string.replace(oldvalue, newvalue)
```

```
File  Edit  Format  Run  Options  Window  Help
old="My Name is Riya"
new=old.replace("Riya", "Rohan")
print("Old string is :",old)
print("New string is :", new)
```

Figure 15.10: Using the replace() Function

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24
AMD64)] on win32
Type "help", "copyright", "credits" or "lice
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Proc
Old string is : My Name is Riya
New string is : My Name is Rohan
>>>
```

Output

count(): This function returns the number of occurrences of a character or word in a string.

Syntax:

```
int string.count(value)
```

```
File  Edit  Format  Run  Options  Window  Help
quote="Practice makes a man perfect!"
val=quote.count("a")
print(val)
```

Figure 15.11: Using the count() Function

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24
AMD64)] on win32
Type "help", "copyright", "credits" or "lice
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Proc
1
>>>
```

Output

isalpha(): This method checks if all the elements in the string are alphabets. This function will return false if it encounters a digit in the string.

Syntax:

```
boolean string.isalpha()
```

```
File  Edit  Format  Run  Options  Window  Help
quote="abd"
val=quote.isalpha()
print(val)
```

Figure 15.12: Using the isalpha() Function

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2
AMD64)] on win32
Type "help", "copyright", "credits" or "licen
>>>
= RESTART: C:/Users/Mitul/AppData/Local/Progr
True
>>>
```

Output

Activity 12.5

What should be the output of following program?

```
information="My favorite subject is Python, I want to develop an AI application
in Python"
val=information.find('T')
print(val)
if val>1:
    print("Python is a good programming language for AI applications ")
else:
    print("My favorite subject is Python")
```

isalnum(): This function returns 'True' if all the elements of the string are alphanumeric, otherwise it returns 'False'.

Alphanumeric is a combination of characters and numeric values. For example, a vehicle registration number, like TN 01 8765, is alphanumeric.

Syntax:

```
booleanstring.isalnum()
```

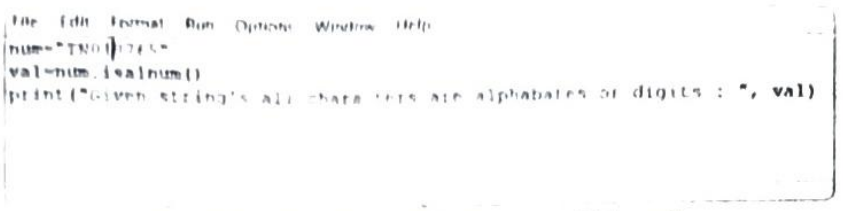
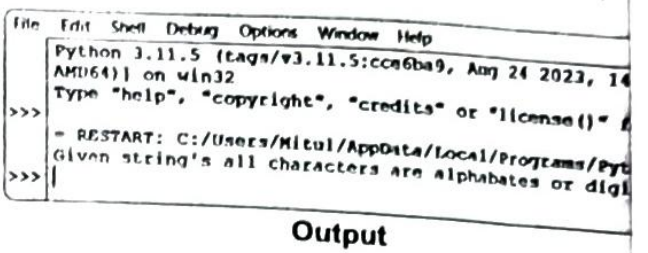


Figure 15.13: Using the isalnum() Function



Output

isdigit(): This method returns True if all the characters in string are digit, it returns False otherwise

Syntax:

```
boolean string.isdigit()
```

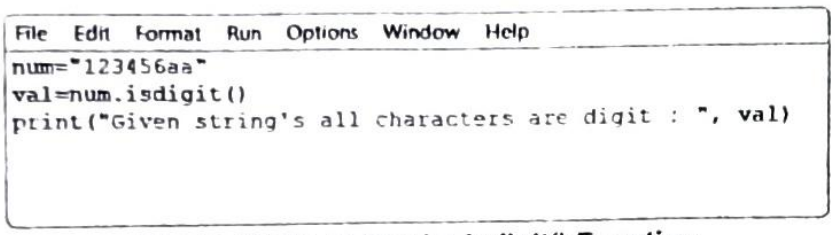
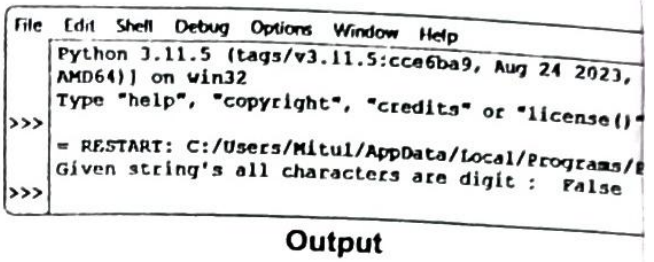


Figure 15.14: Using the isdigit() Function



Output

join(): This method takes all items in an iterable and joins them into one string. A string must b specified as the separator.

Syntax:

```
string string.join(iterable)
```

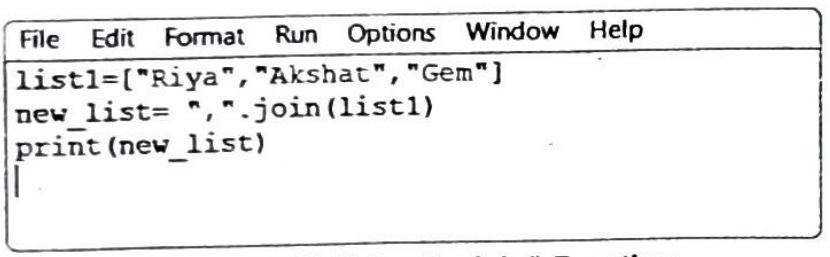
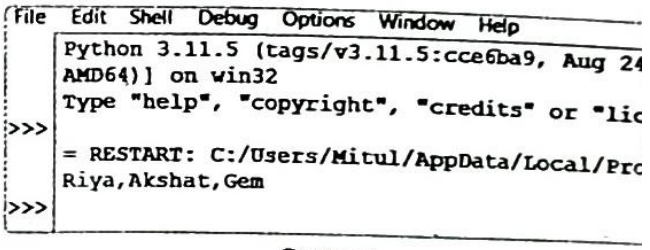


Figure 15.15: Using the join() Function



Output

string[]: It is used for string slicing. A slice is a portion or substring of a string. Selecting a slic is like selecting a character from a string. The slice operator [n:m] is used to return the substrin of the string, beginning with the character at index n and continuing up to the character a position m-1.

Syntax:

```
string string[start:end:step]
```

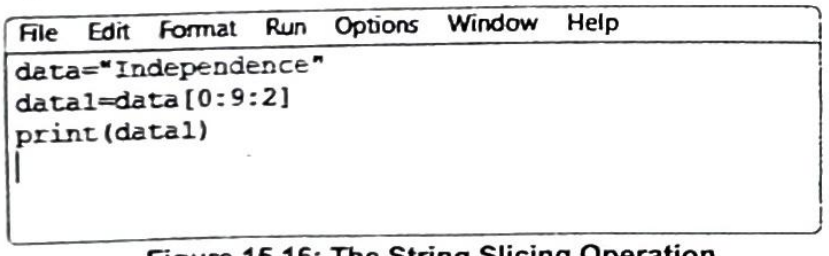
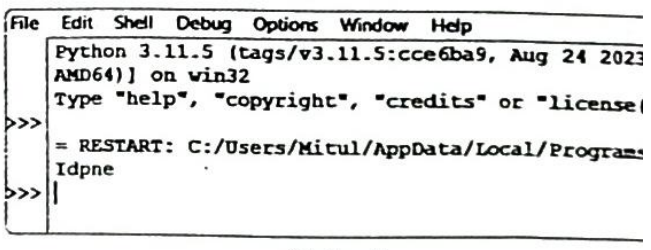


Figure 15.16: The String Slicing Operation



Output

In the above code, the string elements will be taken from the index values 0 to 9 (i.e., Independent with the step value 2, which means that alternate elements will be printed.

Challenge Zone

Find the output of the following program:

```
data="I am very happy"
data1= data[1:4]
print(data1)
data2=data[-4:-1]
print(data2)
```

Output: am
app

ReCap



- String is a collection of characters.
- In python, the strings are declared by using single quotes or double quotes.
- Escape character backslash \ is used to insert characters that are not allowed in a string.
- There are many operators like + and * that have some specific meaning in strings.
- There are many in-built functions in string to support various tasks like len, find, join, isdigit, isalnum, count, islower, isupper, replace etc.

Brain Developer

Section A: Solved Questions

A. State True or False.

1. Only single quotes can be used to declare a string in Python.
2. Blank spaces are counted as character in string.
3. Special characters cannot be used in strings.
4. Backslash (\) is an escape character.
5. The operator is also called as replication operator in string.
6. Triple quotes are used to create multiline strings.
7. String in Python does not allow membership operator.
8. The escape sequence is used to produce a blank space.
9. String slicing means to create a sub string.
10. The tolower() method is used to convert all letters of a string into small letters.

B. Select the correct option.

1. The operator used to join two or more strings is ..... .